Design an AI-supported unit

AI · Level 4 · Lesson 4



Today we will

- Collect data about real school days in a table.
- Build a decision tree of yes/no questions from our data.
- Test our tree and say when it would be wrong.

Remember

- A decision tree decides through a chain of questions.
- A good question splits the days into clearer groups.
- A tree built from a few days can be wrong about new days, and we can see why.

Decision tree from our data

1. Warm-up (5 min): Ask what decides whether we play outside at break.
2. Data (8 min): Fill a table of the last ten days: weather, wind, played outside.
3. Build (14 min): Choose the question that best splits “played” from “stayed in”; repeat on each branch.
4. Test (8 min): Predict today and two made-up days with the tree; check with the class.
5. Wrap-up (5 min): Ask when the tree would be wrong, and what data would improve it.

Prepare a table of about ten days in case the class can’t remember them; real days are better, but invented ones work. Keep two or three features with simple values. At each branch, count the days that played and stayed in: the best question makes each side as unmixed as possible. If two days have the same weather and wind but different answers, no question can split them: call it a day the tree gets wrong, and ask what else was different (an exam, the teacher on duty).

Let’s discuss

1. Which question did our tree ask first, and why?
2. When would our tree give the wrong answer?
3. Is a model you can read easier to trust than one you can’t? Why?

Exit ticket

- One question our tree asks, and one day when it would be wrong.